COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	August 25, 2026	**DATE SUBMITTED:**	August 25, 2026
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ACTIVITY NO. 4.1 **Ansible Basics**

1. PROCEDURE OUTPUT

```
jbbpios@workstation:~/Desktop$ cd CPE212_Pios
jbbpios@workstation:~/Desktop/CPE212_Pios$ git pull
Already up to date.
jbbpios@workstation:~/Desktop/CPE212_Pios$
```

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

```
 GNU nano 6.2                                  inventory.yaml
all:
  children:

    ubuntu:
      vars:
       ansible_user: jp
      hosts:
        server1:
          ansible_host: 192.168.56.101
          ansible_user: jppios
          os_type: Ubuntu

        server2:
          ansible_host: 192.168.56.102
          ansible_user: jpbpios
          os_type: Ubuntu

    centos:
      hosts:
        server3:
          ansible_host: 192.168.56.104
          ansible_user: jpios
          os_type: CentOS


    linuxservers:
       children:
         ubuntu:
                    [ Bad lock file is ignored: ./.inventory.yaml.swp ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
```

```
  ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0
  ignored=0

jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

```
 GNU nano 6.2                                  php_supp.yml *
---
- hosts: all
  become: true
  tasks:

  - name: update repository index
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache2 package
    apt:
      name: apache2
    when: ansible_distribution == "Ubuntu"

  - name: add PHP support for apache
    apt:
      name: libapache2-mod-php
    when: ansible_distribution == "Ubuntu"
Save file under DIFFERENT NAME?
 Y Yes
 N No           ^C Cancel
```

```
jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass install_apac
he.yml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
[WARNING]: Platform linux on host server3 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server3]
[WARNING]: Platform linux on host server2 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [install apache2 package] *************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [server3]: FAILED! => {"changed": false, "cmd": "update", "msg": "[Errno 2] No such file or directory: b'update'"
, "rc": 2, "stderr": "", "stderr_lines": [], "stdout": "", "stdout_lines": []}
ok: [server1]
changed: [server2]

PLAY RECAP *********************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$
```

```
jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ nano php_supp.yml
jbbpios@workstation:~/Desktop/CPE212_Pios/ansible-lab$ ansible-playbook -i inventory.yaml --ask-become-pass php_supp.yml
BECOME password:

PLAY [all] ********************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************
[WARNING]: Platform linux on host server2 is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Platform linux on host server3 is using the discovered Python interpreter at /usr/bin/python3.12, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server3]

TASK [update repository index] ************************************************************************************************
skipping: [server3]
changed: [server2]
changed: [server1]

TASK [install apache2 package] ************************************************************************************************
skipping: [server3]
ok: [server2]
ok: [server1]

TASK [add PHP support for apache] *********************************************************************************************
skipping: [server3]
ok: [server1]
changed: [server2]

TASK [update repository index] ************************************************************************************************
skipping: [server2]
```

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab

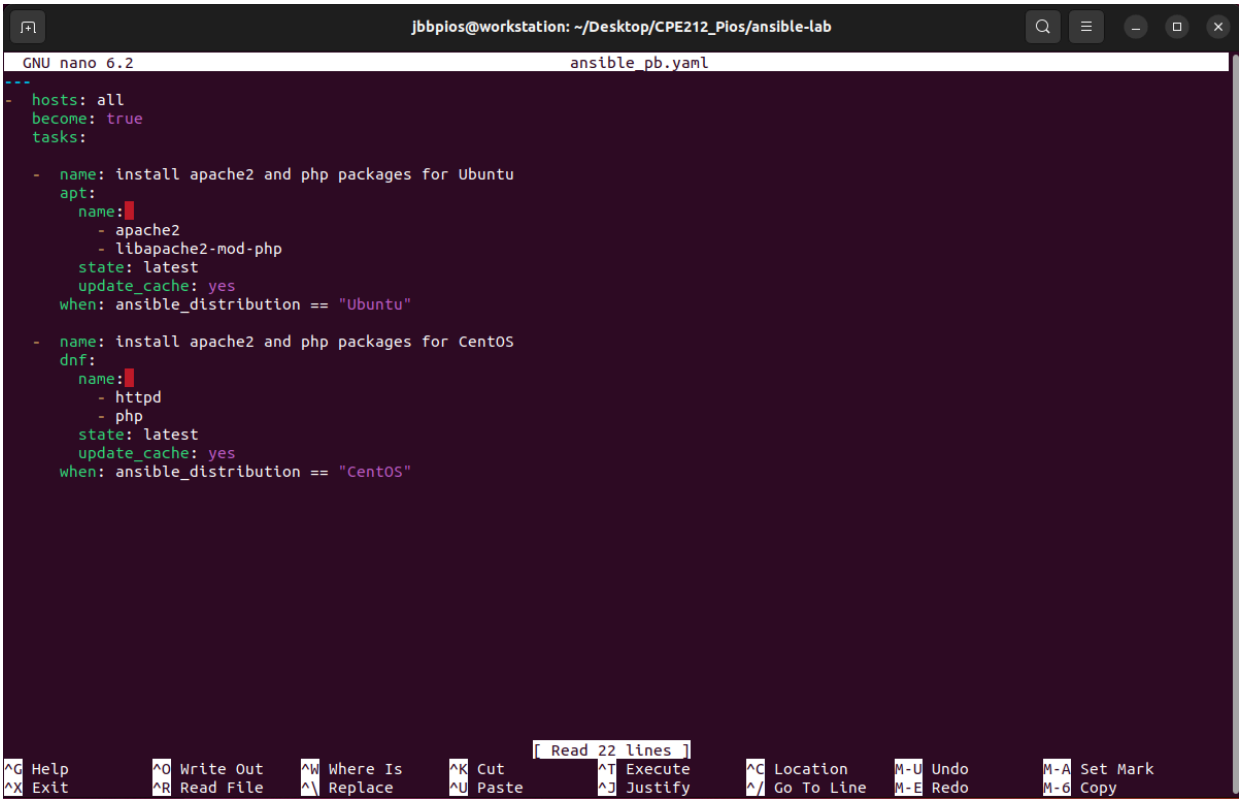
```
 GNU nano 6.2                                        php_supp.yml *
---
-  hosts: all
   become: true
   tasks:

   -  name: update repository index Ubuntu
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

   -  name: install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

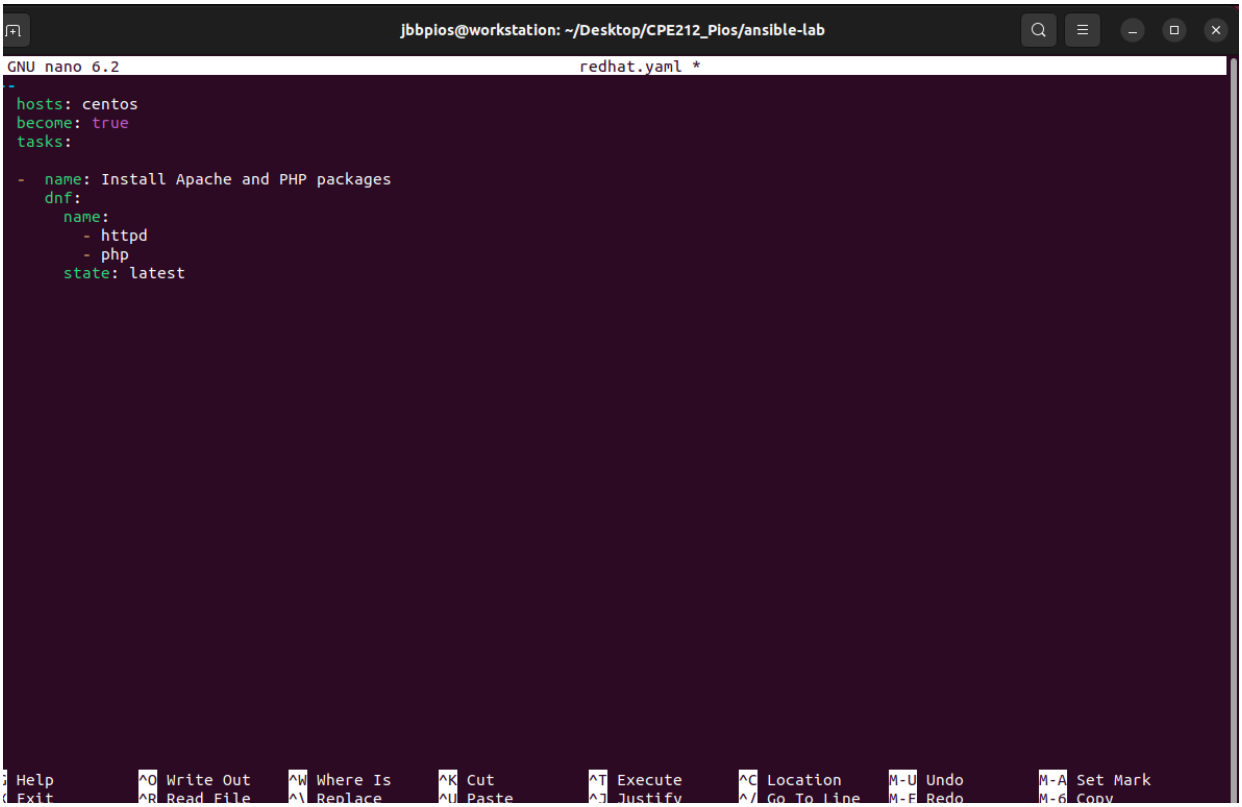
   -  name: install apache2 and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a playbook that could do the previous tasks in Red Hat OS.



Reflections:

Answer the following

1. Why do you think refactoring of playbook codes is important?

Refactoring playbook codes is important because it makes the playbook cleaner, shorter, and easier to understand. It also reduces repetitive tasks and makes it easier to manage different operating systems while keeping the same functionality.

2. When do we use the when command in playbook?

We use the when command when a task should only run under a specific condition. In this activity, it was used to make sure the appropriate tasks run depending on whether the target machine is using Ubuntu or CentOS.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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